

Demystifying Index Rebalancing: An Analysis of the Costs of Liquidity Provision

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KEY FINDINGS

- Today, virtually all stocks are part of some index, so the conventional approach to examining index adds and deletes needs to be modified to account for flows by looking across indexes at weight changes scaled by the AUM tracking each index.
- Liquidity cost is measured by the temporary price impact, that is, the negative of the return reversal following the effective date; permanent impacts are not actionable.
- Measured this way, the cost of liquidity provision is small overall. This is consistent with a highly competitive market for liquidity provision.

ABSTRACT

Recent academic studies claim that price impacts associated with index rebalancing can be large and may represent a hidden cost to fund investors. The authors examine the return behavior around 18,871 rebalances for the Russell indexes from 2016–2021. A key difference from the literature is that they estimate net flows for each stock aggregated across all Russell indexes rather than the gross flows into each index. This is important because stocks migrating from one index to another may see positive or negative flows depending on the assets tracking those indexes. They find many instances in which the costs of liquidity provision, as measured by the beta-adjusted temporary impact, are negative. This is consistent with a highly competitive market for liquidity provision.

The enormous growth of assets in exchange-traded funds (ETFs) and index funds has given rise to concerns that periodic rebalancing flows from these products may result in large volumes and price movements in stocks being added or deleted from an index.¹ All indexes, including market-capitalization-weighted indexes, need to periodically reconstitute and rebalance due to changes in index inclusion criteria or corporate actions.² This is true even of market-cap-weighted indexes that are naturally self-rebalancing, reducing turnover and transaction costs relative to alternative weighting schema.³ In theory, if an investor owns the whole market (i.e., all stocks),

¹ Rebalancing flows may also arise from nonindex funds including closet indexers and active managers who may add to price movements associated with rebalancing as they seek to reduce tracking error.

² Reconstitution refers to adding or deleting securities from an index based on the index criteria. Rebalancing refers to adjusting index weights as specified by the index provider to reflect corporate actions such as mergers, spin-offs, and so on.

³ Many traditional equity indexes were originally intended as performance benchmarks for the entire market or major market segments and not necessarily as investment vehicles in that they ignore

migrations from one segment to another should have minimal impacts to the investor's overall portfolio. But many investors own segments of the market, not the entire market in contrast to Sharpe (1964), and hence, the migration of stocks from one index to another may have return effects arising from market impact.

Recent academic research claims that index/ETF rebalancing, typically at the close on the effective date of the addition or deletion, can give rise to large price effects that represent a hidden cost to index investors that does not show up because their tracking error to an index is minimal.⁴ This literature extends a long line of older research going back decades that has studied the effects of index additions and deletions, mostly focusing on the S&P 500 index.⁵ That approach was sensible when indexing was nascent, and one could examine price impacts without consideration of other rebalancing flows around the same dates. Today, most US stocks that trade daily are members of several indexes, meaning that the price impacts associated with index rebalancing and reconstitution can no longer be examined in isolation. Further, the market for liquidity provision is highly competitive compared with the 1980s, as we discuss in the following.

This article discusses the practical realities of index rebalancing using a large database of 18,871 rebalances of the Russell indexes from 2016 to 2021. The Russell indexes are major benchmarks for index managers, and the price effects around these rebalances are of considerable interest to financial economists (e.g., see Madhavan (2003), Chinco and Sammon 2022, and Micheli and Neuman 2020) because Russell index weight changes are based on observable market capitalization.⁶ Consequently, liquidity providers have ample time to position themselves prior to the announcement and effective dates.

An important difference from the earlier literature is that we estimate net flows for each stock aggregated across all Russell indexes rather than the gross flows per index. By doing so, we determine whether each stock is a net buy or net sell among all index investors participating in the reconstitution. The reason this factor is so important is that the assets benchmarked to different indexes affect the magnitude and direction of price impact. For example, suppose a stock in a small-cap index B is promoted from B (in which it had a high weight) to a high-cap index A (in which it now enters with a low weight). The promotion (i.e., addition to A and deletion from B) may, depending on the assets tracking each index and the weight changes, be associated with either net selling or net buying. We focus on the Russell indexes because they are all rebalanced on the same day, the last Friday in June, allowing us to compute the net flows across all indexes. Additionally, no other major index provider rebalances in that week.

We estimate both permanent and temporary price impacts, the former reflecting the information conveyed by index changes, and the latter representing the true cost of liquidity provision. We also appropriately beta adjust returns for market movements to reduce the noise associated with estimation. We estimate beta prior to the

transaction costs, the need for cash, and so on. These indexes typically use market capitalization weighting schema. Newer indexes, including some thematic and factor indexes, are now explicitly designed for investment and may use alternative weighting methodologies.

⁴See Li (2022); Reilly (2022); Edelen, Evans, and Kadlec (2013); and Petajisto (2011). The financial press has also picked up on this idea (see, e.g., Johnson 2022).

⁵See Harris and Gurel (1986); Jain (1987); Lynch and Mendenhall (1997); Strongin, Petsch, and Sharenow (2000); and Madhavan and Ming (2003), among others.

⁶Ben-David, Franzoni, and Moussawi (2018) used the Russell ranking process to study the impact of index inclusion on the volatility and liquidity of constituents. Other studies (see Wei and Young 2019) used the annual reconstitution of the Russell 1000 and 2000 Indexes to study the effect of exogenous changes institutional ownership on firm outcomes. See also Schmidt and Fahlenbrach (2017) and Appel, Gormley, and Keim (2019) whose focus was governance.

announcement day as subsequent market movements arising from information and momentum effects may change beta.⁷ We find that information effects are pervasive and are positive for rebalancing events with positive flows and negative otherwise. Measured correctly, we find many instances where costs (measured by the transitory component of market impact) are negative. Across both buys and sells, the overall cost of liquidity provision is very low. This is consistent with a highly competitive market for liquidity provision.

PRACTICAL REALITIES OF INDEX REBALANCING

Market Participants

We begin with a discussion the complexity and interdependencies in the ecosystem around rebalancing and the nature of competition for liquidity provision. There are several key players in the ecosystem that can be categorized into four groups: index providers, index asset managers, rebalance market facilitators, and liquidity providers. Index providers⁸ are institutions that create, calculate, and maintain market indexes. Some of the well-known index providers include FTSE Russell, MSCI, and SPDJ. In recent years we have seen a proliferation of index providers creating niche market indexes particularly in the thematic and Environmental, Social, and Governance (ESG) space such as Qontigo/Stoxx, Solactive, and Morningstar. Index asset managers are responsible for managing the funds with the primary goal of replicating the returns of the portfolio's benchmark. As index assets continue to grow, these managers have become critical to the rebalance ecosystem and the associated price movements. BlackRock, Vanguard, and State Street are some of the largest index asset managers. Index managers are at the forefront of the start of the rebalance process bringing reality to the changes announced by the index providers.

Rebalance facilitators, as described by its group name, are market participants that facilitate the success of an index rebalance process. Participants include various arms of a brokerage firm (J.P. Morgan, Citi, Morgan Stanley) providing trading strategies and access to capital and liquidity and custody firms (BNY Mellon) that ensure the deposit and delivery of securities affected in the rebalance. Liquidity providers complete the picture of the rebalance ecosystem offering to take the other side of the rebalance trade presented by the index managers. Liquidity providers include other market participants such as hedge funds, active fund managers, broker dealers, and corporations. The last point is important because corporations are often ignored as sources of liquidity but are increasingly playing a key role in rebalancing as we discuss below.

TIMELINE OF EVENTS

In what follows, we provide a detailed discussion of the elements around the annual reconstitution of the FTSE Russell indexes on the last Friday of June of each year. The fact that all Russell indexes rebalance on the same day is convenient for our examination of net flows across indexes. We also note that there are no other major rebalances from other index providers that week. Although our discussion is specific to this index provider, this process is very similar for other indexes too.

⁷We estimate 52-week betas from Bloomberg ending on the May ranking date and Winsorize at ± 3 sigma to calculate the beta-adjusted returns.

⁸See An, Benetton, and Song (2022).

Pricing in the Preannouncement Period: March–May

During the preannouncement period, index providers are preparing the new index universe by recreating the universe from scratch. This process includes incorporating security regulatory changes if applicable, updating security shares outstanding and floated shares, updating financial metrics, and establishing market price dates for the new universe. While this is taking place, rebalance facilitators, such as index researchers, both at brokerage firms, are using the more recent index methodology and publicly available market information to predict anticipated changes to the index to estimate the size of the upcoming index rebalance. Liquidity providers, such as hedge funds, use the index predictions to establish trade positions (typically additions and deletions) in anticipation of supplying liquidity on the rebalance effective date. To facilitate their positions, they seek capital and borrow securities through broker dealers (prime brokerage and securities lending division).

Announcement to Effective Period (June)

Once the index providers announce the actual index changes: additions, deletions, migrations, shares, and float updates, the participants in the ecosystem actively position themselves as described in the following:

- Index providers: Respond to inquiries related to new universe, calculations, and provide index updates as necessary.
- Index managers: Review announced index changes from index provider, assess holdings threshold and restrictions, secure cash for trade execution, inquire about portfolios, interact with broker dealers on liquidity prospects, and strategize on securities that require more execution attention. Establish trades for event day.
- Rebalance facilitators:
 - Broker dealers: A variety of work and preparation is done with the brokerage firms. The index researchers publish announced index changes, interact with clients (portfolio managers, analysts, company investor relations) on market environment; also, they reconcile predictive versus announced results. The portfolio trading desk interacts with index managers as the execution team for their trades and to secure balance sheet allowance for index principal bids and custom in-kind baskets creates and redeems.
 - ETF authorized participants: Prepare for ETF underlying changes due to the rebalance to ensure proper creation/redemption transactions.
 - Securities lending desks: Review announced results to adjust for potential recalls from security owners; source locates for more lending activity.
 - Prime brokerage: The financing arm secures cheap lending rates for their clients, such as hedge funds and quant funds, looking to participate in the rebalance.
 - Delta one: Establish trades to rebalance underlying index swaps to maintain tracking of underlying index. Activity may vary depending on the counterparty risk.
- Liquidity providers:
 - Hedge funds (index arbitrageurs): Reconcile predictive results with announced results. Engage in unwinding trades that have either worked well (profit taking) or performed poorly (cut losses). Establish new trades ahead of event day to provide liquidity.
 - Active long-only managers: Review Index rebalance results to source for liquidity opportunities either to leg into a trade or exit a trade.

- Corporations: Provide liquidity by issuing a secondary offering if it is a new addition to the index (Facebook added to the S&P 500 in 2013). Remove excessive selling of security due to stock deletion by issuing a Dutch tender (Yahoo's reorganization from corporation to 40-Act).
- Buyback programs: Seek opportunities to buy back stock for corporations during liquidity events.

Post Effective/Rebalance Period: End of June

Like the anticipation of a major event such as the Super Bowl, the effective day of the Russell reconstitution brings all interested parties to the market. Residual work continues even after effective date. Index managers complete rebalance trade by managing execution of highly illiquid securities; liquidity providers (hedge funds) establish reversal trades to capture rebalance induced price movements; and broker dealers (portfolio trading) manage execution of residual trades from event day/ establish reversals.

This summary highlights the complexity of the ecosystem around rebalances today including the role of nontraditional sources of liquidity such as corporations.⁹ Essentially, the rebalance effective date provides a focal point for the ecosystem players to come together as in an auction at the close, as described in Admati and Pfliederer (1988) and Madhavan (1992).

EMPIRICAL ANALYSIS

Definitions of Price Impact

Let $p_{i,t}$ denote the price (adjusted for dividends and corporate actions such as splits, etc.) of a security $i = 1, \dots, N$ that is being rebalanced in an index as of date $t \in \{a, e, d\}$. The three dates $\{a, e, d\}$ correspond to, respectively, the closing date just prior to the security rebalance announcement, the date on which the security is effectively in or out of the index, and an end date after the rebalance is over. In our empirical work, we typically select $d = e + 3$ because we find that excess volumes and volatility return to normal by this date. We define the total continuously compounded or log return¹⁰ from just prior to the announcement to the close on the effective date as

$$r_i = \ln \left(\frac{p_{i,e}}{p_{i,a}} \right) = \ln(p_{i,e}) - \ln(p_{i,a}) \quad (1)$$

We can then break the total return into permanent and temporary impacts as discussed in Keim and Madhavan (1997):

$$\pi_i = \ln \left(\frac{p_{i,d}}{p_{i,a}} \right) = \ln(p_{i,d}) - \ln(p_{i,a}) \quad (2)$$

⁹An example of the importance of periodic rebalancing is at: <https://www.morningstar.com/articles/960641/the-strange-and-happy-tale-of-voya-corporate-leaders-trust> which describes a fund inception in November 1935 with 30 original constituents. The fund was not permitted to trade, so the remaining 20 constituents reflect corporate actions since inception, resulting in a weighting scheme dominated by its largest holding (43.3%) as of February 2021.

¹⁰Continuously compounded returns have an attractive additive feature and can readily be converted to standard price returns by taking the exponent of the log return and subtracting 1. Over the short intervals we focus on here, there is very little difference between the two return definitions.

$$\tau_i = \ln\left(\frac{p_{i,e}}{p_{i,d}}\right) = -(\ln(p_{i,d}) - \ln(p_{i,e})) \quad (3)$$

In Equation 2, the permanent impact is defined as the return from the announcement date to a date post the rebalance when volumes and volatility have returned to normal. A stock being added to a major index that experiences positive flows is likely to have positive permanent impact (i.e., $p_{i,d} > p_{i,a}$) because liquidity increases and the stock is easier to hedge, reducing spreads. Additions to a major index also implies an improvement in information quality with more analyst coverage. The opposite is true of deletions from major indexes.

In Equation 3, the temporary or transitory price impact is the negative of the log price return after the effective date. The older literature on index adds and deletes suggests that a stock that is added to an index will have a higher price at the close on the effective date than after that time so that $p_{i,e} > p_{i,d}$. The reverse is the case for a deletion. To the extent that liquidity suppliers seek a premium to provide liquidity on the effective date, that price premium should reverse out in the subsequent few days. We see from Equations 2 and 3 that the sum of the two components is the total return in Equation 1, that is, $r_i = \pi_i + \tau_i$.

Observe in Equations 1–3 that we do not sign returns based on whether a security is added or deleted from an index as is common in the previous literature. In reality, flows that occur on index rebalance dates when index weight changes cannot be considered in isolation of each other. Consider, for example, a large-cap stock that experiences an increase in weight in the Russell 1000 Growth index at the same time its weight decreases in the Russell 1000 Value index. The price impact associated with this change is a function of the net flows for these directionally opposed trades. Essentially, with the proliferation of indexes (Madhavan 2016) and growth of index assets, most stocks are part of several indexes. Hence, the dollar flow for each stock across all indexes at a point in time is paramount to determining price impact.

To formalize this, let $A_{j,t}$ denote the total assets (in dollars) tracking index $j = 1, \dots, J$ at a point in time t when index rebalances are announced. Further, denote by $w_{i,j,t}^-$ the weight of security i in index j at time t prior to the rebalance, and let $w_{i,j,t}^+$ denote the corresponding weight immediately after the rebalance announcement. Then the projected money flow into security i at time t across all J indexes is defined as

$$f_{i,t} = \sum_{j=1}^J (w_{i,j,t}^+ - w_{i,j,t}^-) A_{j,t} \quad (4)$$

When the coefficient $w_{i,j,t}^+ - w_{i,j,t}^-$ is positive (negative), this corresponds to an upweight (downweight) of security i in index j . Special cases are $w_{i,j,t}^- = 0$ for index adds and $w_{i,j,t}^+ = 0$ for deletes. Net buys and sells are determined by the positivity of the expression in Equation 4. We define a trade indicator variable $x_{i,t} = \text{sign}(f_{i,t})$. So, $x_{i,t} = 1$ for a buy (i.e., when $f_{i,t} > 0$) and $x_{i,t} = -1$ for a sell when $f_{i,t} < 0$. With these definitions in place, we now turn to our data and methodology.

Data and Methodology

To understand the stock dynamics around Russell reconstitution events, we reconstruct the complete Russell trade for 2016 to 2021. By complete trade, we mean the estimated net flows for each stock aggregated across all Russell indexes rather than the standalone, gross flows per index. By doing so, we determine whether each stock is a net buy or net sell among all indexers participating in the reconstitution.

EXHIBIT 1**Estimated Assets Following Russell Indexes and Turnover**

Index	AUM (\$ billion)	Turnover (%)	Turnover (\$ billion)
Russell 1000	482.6	2.78	13.44
Russell 3000	430.5	2.63	11.34
Russell 1000 Growth	299.3	13.32	39.87
Russell 2000	297.5	12.40	36.91
Russell 1000 Value	196.9	13.38	26.34
Russell TOP 200	114.3	2.26	2.58
Russell Midcap	55.1	9.38	5.17
Russell Small Cap	53.0	10.26	5.44
Completeness			
Russell 2000 Value	47.3	25.42	12.01
Russell 2000 Growth	45.1	28.44	12.83
Other Russell Indexes	66.9	23.64	15.82
Total	2,088.5	8.70	181.74

SOURCES: BlackRock as of July 31, 2021, based on FTSE Russell surveys. Russell index constituents and IDC pricing provided by Refinitiv via QA Direct.

Classifying each stock as a buy or sell is more nuanced than just tracking its before and after membership in each Russell index because it requires an estimate of both the weight change of every stock in two dozen Russell indexes and the assets under management (AUM) tracking each index. The index weight changes are widely forecasted before the Russell ranking date in May and leading up to the announcement date on the first Friday of June. After the announcement date, at which point the index memberships are known with certainty (save for corporate actions and extraordinary events), the constituent weights after reconstitution depend almost entirely on price changes leading up to the event. Ex-post, historical weight changes can be calculated using end-of-day index constituent data provided by the index vendor through various data channels.

To estimate index tracking AUM, we use FTSE Russell surveys that are published in July of each year. Their survey methodology combines assets from ETFs, retail and institutional mutual funds, separate accounts, and commingled funds from clients who license Russell indexes for fund benchmarks. The reported tracking AUMs are approximations to the

total index assets tracking each index, and FTSE Russell handles missing survey responses by augmenting with publicly available fund AUM data. We further adjust these AUMs by 11-month index returns to handle the time lag between survey reporting and next year's Russell reconstitution. Combining AUM estimates with the index weight changes yields the dollar amount of gross one-way turnover per index, as shown in Exhibit 1. The turnover columns are gross figures with no netting among indexes. Exhibit 1 also shows more turnover (measured one way) on a percentage basis for style indexes (cf., 2.78% for Russell 1000 versus 13.32% and 13.38% for Russell 1000 Growth and Value, respectively). This makes sense because, between index rebalances, style index constituents can migrate on two dimensions of market cap and factor style.¹¹

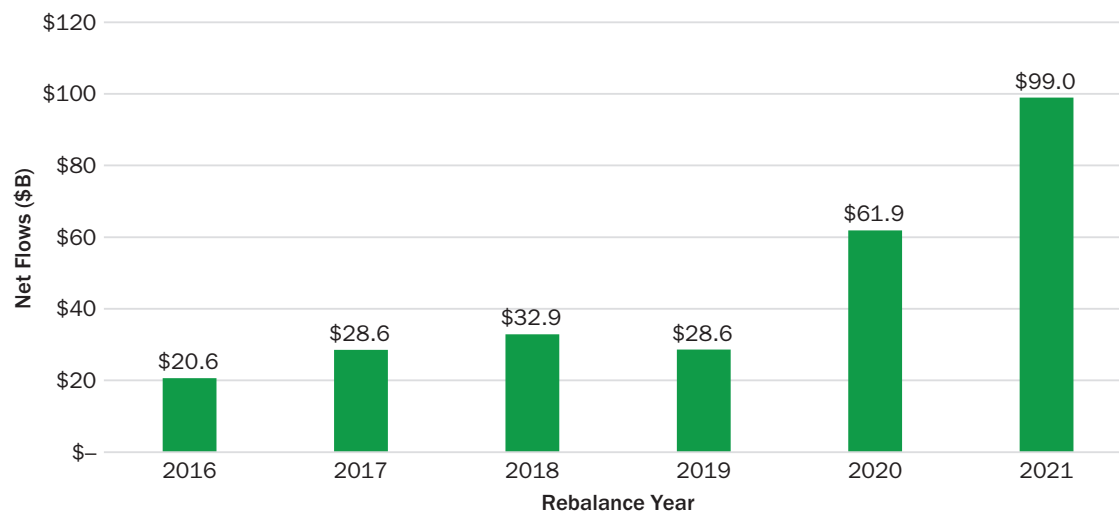
Exhibit 2 shows the net flows, as described in Equation 4, considering every stock's weight in each index and the tracking assets to each index from 2016 to 2021. In all, there are 18,871 stock-level rebalance events in the dataset.¹² In our multivariate analysis, we augment these data with estimates of beta, market capitalization, and average daily dollar volume sourced from Bloomberg and eliminate any records with missing returns data. Exhibit 2 shows a fivefold increase in net dollar flows from 2016–2021 and demonstrates the enormity of trades that occur on what is often the largest trading day of the year. Years in which volatility is high, including the US market recovery from March 2020 COVID-19 lows spanning the June 2020 and June 2021 rebalances, necessitate large index rebalances because of large market cap shifts among index constituents.

¹¹Aked and Moroz (2013) compared turnover in traditional versus fundamental indexes. See also Shankar, Miller, and Balakrishnan (2015) and Levy (2016).

¹²We require that each security belongs to the Russell 3000 either before and/or after the reconstitution date; that is, we do not include microcap stocks that are in the Russell 3000 Extended but not in the Russell 3000.

EXHIBIT 2

Net Flows on Russell Reconstitution Days 2016–2021



SOURCES: BlackRock as of July 31, 2021, based on FTSE Russell surveys. Russell index constituents and IDC pricing provided by Refinitiv via QA Direct.

With regard to reconstitution day execution costs, larger asset managers can internally cross trades that are buys for one tracking fund but sells for another. Although this saves on trading costs for the benefit of the fund client, these trades still *print to the tape*, meaning that their volumes are captured by the exchanges and are included in the total trading volume published for that day. For this reason, implementation costs based purely on daily trading volume, bid–ask spreads, and market impact models are often exaggerated. Also, market impact models that use the ratio of the day’s trading volume to, say, the trailing 20-day average daily volume (ADV) for that stock, are not properly calibrated to consider the excess liquidity that shows up on major index rebalance days.

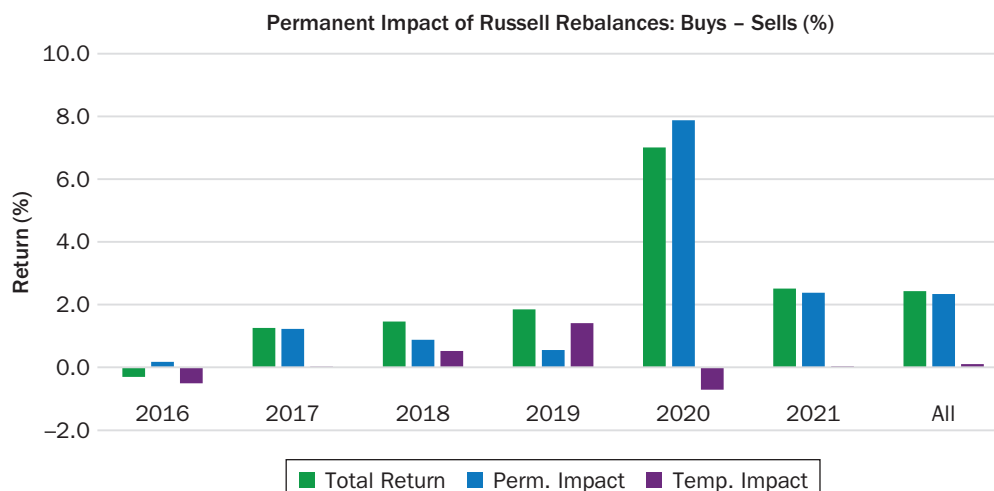
Exhibit 3 shows the decomposition of total returns from time a to time e into the permanent impact from a to d (where $d = e + 3$) and the temporary impact from e to d . Returns are calculated at the stock level, averaged across buy and sell baskets separately, then differenced to represent a dollar-neutral trade that is long the buy basket and short the sell basket.¹³ The exhibit clearly shows that the temporary impact, our metric for the cost of liquidity provision, is small relative to permanent impact and even negative in some years.

Exhibit 4 shows a refinement of the data in Exhibit 3 into buy and sell categories (Panels A and B) along with the counts of the number of securities by year in each category. As neither the buy nor sell sides of the trade should be viewed in isolation, Panel C shows the buy minus sell difference portfolio returns and impacts from forming a dollar neutral, hedged portfolio. Exhibit 5 augments the data in Exhibit 4 with additional data across all years, including the median impact, standard error of the mean, and the standard deviation of impacts. We see that the means are both statistically and economically significant across buys and sells. Median total impacts for buys are much smaller than the mean, but this is not the case for sells. The standard deviations are indicative of high variability in impacts across events and over time.

¹³This essentially signs the trade based on the direction of flow. The exhibit is for illustrative purposes only as it may not be possible to short the sell basket.

EXHIBIT 3

Total, Permanent, and Temporary Price Impact: 2016–2021



NOTES: We report three price impact measures for a dollar neutral portfolio of buys minus sells as defined in Equations 1–3. Past performance does not guarantee future results. It is not possible to invest directly in an unmanaged index.

SOURCES: BlackRock as of January 31, 2021, based on FTSE Russell surveys. Russell index constituents and IDC pricing by Refinitiv via QA Direct.

Exhibit 6 shows the permanent and temporary beta-adjusted returns across all events based on a decile sort flow scaled by the 20-day average daily dollar volume (ADV) prior to announcement. We estimate each stock's beta using data prior to the announcement day to avoid picking up any effects associated with index changes. There are approximately 1,887 events in each decile. Recall from Equations 1–3 that the return components are not signed. We see a strong relation between the total impact and signed flow. Given the large overlap in time between *a* and *e* and between *a* and *d*, it is not surprising that this pattern holds.

Exhibit 6 highlights the extreme variation in scaled flow in the smallest and largest deciles. For example, in the top decile, the scaled flow is over 20.3 days of daily volume. Not surprisingly, the relation of scaled flow to market capitalization is nonlinear; the extremes of negative and positive scaled flows are in smaller-cap names and least in the largest-cap names. Exhibit 6 also shows little evidence that beta matters. Of considerable interest, the permanent effects are clearly related to the flow variable, but the temporary impacts are not. Indeed, the temporary effect is positive 1.31% in the highest net outflow decile.

Categorization by Transition Type

In this section, we investigate the price impacts by transition type. For the Russell 1000, we categorize transitions as (a) additions, coming into the Russell 1000 from outside the Russell 3000; (b) promotions, coming from a stock moving from the Russell 2000 into the Russell 1000; (c) stays, where the stock remains in the Russell 1000; (d) demotions, in which a stock previously in the Russell 1000 is moved to the Russell 2000; and (e) deletions, in which the stock that was in the Russell 1000 is moved out of the Russell 3000.¹⁴ Correspondingly, for the Russell 2000, we define: (f) additions, coming from outside the Russell 3000; (g) stays, in which the stock

¹⁴ Recall that the Russell 3000 is the union of the Russell 1000 and the Russell 2000.

EXHIBIT 4**Price Impacts (percentage) by Buy/Sell Category and by Year**

	Count	Total Return	Perm. Impact	Temp. Impact
Panel A: Buys				
2016	1,914	-0.07	-1.19	1.15
2017	1,398	1.54	2.31	-0.83
2018	2,010	2.63	1.99	0.53
2019	2,077	2.10	0.16	1.84
2020	1,592	1.09	0.47	0.58
2021	1,057	2.16	0.34	1.67
All	10,048	1.56	0.63	0.86
Panel B: Sells				
2016	1,210	0.24	-1.37	1.66
2017	1,700	0.27	1.08	-0.86
2018	1,092	1.17	1.10	0.01
2019	1,039	0.25	-0.39	0.43
2020	1,549	-5.92	-7.41	1.28
2021	2,233	-0.35	-2.04	1.63
All	8,823	-0.87	-1.70	0.75
Panel C: Buys Less Sells				
2016	3,124	-0.30	0.18	-0.51
2017	3,098	1.26	1.23	0.03
2018	3,102	1.46	0.89	0.53
2019	3,116	1.85	0.55	1.41
2020	3,141	7.01	7.88	-0.71
2021	3,290	2.51	2.38	0.04
All	18,871	2.43	2.34	0.11

NOTES: We report three price impact measures as defined in Equations 1–3 in percentages and in Panel C for a dollar-neutral portfolio of buys minus sells in percentage points. We categorize rebalances as buys versus sells based on the sign of Equation 4. Past performance does not guarantee future results. It is not possible to invest directly in an unmanaged index.

SOURCES: BlackRock as of July 31, 2021. Russell index constituents and IDC pricing via Refinitiv QA Direct.

stays in the Russell 2000; and (h) deletions, in which the stock is deleted from the Russell 2000.

Exhibit 7 shows the (beta-adjusted) mean permanent and temporary impacts in percentages, cut by trade sign and transition category for the Russell 1000 in Panel A and the Russell 2000 in Panel B, along with the count of events.

We observe that most of the events are stays in their index. For the Russell 1000 in Panel A, we observe that all 72 pure additions were positive flow events, and all 10 deletions were negative flow events. Demotions from the Russell 1000 were associated with positive cash flows because these stocks had a larger weight in the Russell 2000. Most promotions are associated with negative flows and only seven with positive flows. For the Russell 2000, all additions were associated with positive flows and all deletions with negative flows. The permanent impacts tend to go the way we would expect given the flows except for the seven Russell 1000 promotions, likely a small sample artifact. The temporary impacts, however, are generally positive, even for cases when there is negative flow in which we would expect that the price three days after the effective date would be higher than the effective date price implying a negative return. As a result, across all events, the mean temporary impact is relatively small.

Regression Analysis of Price Impact

We can better understand the relation between the price impacts and net demand via regression analysis. In a classic microstructure model, price impacts (r_{it} , π_{it} , τ_{it}) are modeled as a function of trade direction $x_{i,t}$ and scaled order flow $f_{i,t}$. For total impact, for example, we model the impact for stock i at time t as

$$r_{i,t} = \lambda_0 + \lambda_1 x_{i,t} + \lambda_2 f_{i,t} + \varepsilon_{it} \quad (5)$$

and similarly for the permanent and temporary impacts. In Equation 5, ε_{it} is the error term, λ_1 captures

the half-spread, and λ_2 is the price impact coefficient (Kyle 1985) that captures the effect on price of the scaled flow. Note that the independent variables are signed, so we expect that $\lambda_1 > 0$ and $\lambda_2 > 0$.¹⁵ Intuitively, there is a fixed effect from trade direction and a further price movement as liquidity is consumed directionally.

Exhibit 8 reports regressions for total, permanent, and temporary price impacts, all beta adjusted and measured in percentages, across 18,871 rebalance events in the period 2016–2021 on trade sign and scaled flow, defined as the signed dollar

¹⁵ Although trade sign is derived from the sign of flow, leading to a concern over multicollinearity, the correlation of the two variables is low, 0.247. Multicollinearity does not affect the coefficient estimates but can increase the individual standard errors.

EXHIBIT 5**Price Impacts (percentage) by Buy/Sell Category: Means and Medians**

	Buys (N = 10,048)			Sells (N = 8,823)		
	Total	Permanent	Temporary	Total	Permanent	Temporary
Median	0.64%	-0.11%	0.74%	-1.11%	-1.81%	0.69%
Mean	1.56%	0.63%	0.86%	-0.87%	-1.70%	0.75%
S.E. mean	0.12%	0.13%	0.05%	0.11%	0.12%	0.06%
Std. Dev	11.81%	13.20%	5.40%	10.60%	11.53%	5.70%

NOTES: We report three price impact measures based on Equations 1–3 in percentages. Past performance does not guarantee future results. It is not possible to invest directly in an unmanaged index.

SOURCES: BlackRock as of July 31, 2021. Russell index constituents and IDC pricing via Refinitiv QA Direct.

EXHIBIT 6**Sample Means Based on a Decile Sort of Flow/ADV**

Decile	Flow/ADV	Market Cap (\$million)	Beta	Total (%)	Perm (%)	Temp (%)
1	-10.73	2,306	1.11	0.07	-1.29	1.31
2	-0.70	10,034	1.12	-0.86	-1.87	0.95
3	-0.29	19,829	1.12	-1.22	-1.99	0.71
4	-0.13	21,709	1.11	-1.44	-1.99	0.36
5	-0.02	19,974	1.10	-0.63	-1.07	0.38
6	0.10	16,850	1.12	0.70	-0.13	0.77
7	0.29	9,579	1.11	0.82	-0.11	0.86
8	0.68	4,184	1.08	1.66	0.71	0.86
9	1.86	1,991	1.08	1.82	0.81	0.93
10	20.28	1,155	1.01	3.33	2.40	0.91

NOTES: Based on a decile sort of scaled flow in which flow (in dollars) is based on Equation 4 and divided by 20-day average daily dollar volume prior to the announcement date. We report three price impact measures as defined in Equations 1–3 and expressed in percentages. Past performance does not guarantee future results. It is not possible to invest directly in an unmanaged index.

SOURCES: BlackRock as of July 31, 2021. Data on market capitalization (in millions of dollars), beta, and ADV are from Bloomberg.

flow from Equation 4 divided by 20-day average daily dollar volume in the stock. The number of observations (degrees of freedom) may be different for each impact as some measures may be missing over the window of estimation.

All estimated flow coefficients in Exhibit 8 are statistically significant, with the highest significance levels for total and permanent impacts. Trade sign is economically and statistically significant for total and permanent impact but not for temporary impact. The low *R*-squared in all three regressions leads us to conclude that many other factors beyond rebalancing flows affect stock returns, even in the narrow three-day window from the effective date. The estimates in Exhibit 8 are also robust to the use of non-beta-adjusted returns and different definitions using a shorter window around the effective date of $T - 5$ to $T + 3$. We also estimated fixed-effects models using the transition categories as shown in Exhibit 7. This marginally improved the *R*-squared but did not alter our conclusions in any material way and are not reported here.

EXHIBIT 7**Mean Permanent and Temporary Price Impacts (percent) by Flow Sign and Transition Category**

	Positive Flow			Negative Flow		
	Count	Permanent Impact (%)	Temporary Impact (%)	Count	Permanent Impact (%)	Temporary Impact (%)
Panel A: Russell 1000						
Additions	72	5.48	-0.48			
Promotions	7	-3.38	0.34	207	0.45	-0.04
Stays	2,049	0.36	0.43	3,670	-1.61	0.51
Demotions	237	-3.49	2.20			
Deletions				10	-7.43	1.40
Panel B: Russell 2000						
Additions	1,107	2.17	0.85			
Stays	6,576	0.56	0.96	4,058	-1.90	0.90
Deletions				878	-1.66	1.26
Overall	10,048	0.63	0.86	8,823	-1.70	0.75

NOTE: We report three price impact measures as defined in Equations 1–3, beta adjusted, and expressed in percentages.

SOURCE: BlackRock as of July 31, 2021, in which flow (in dollars) is based on Equation 4.

EXHIBIT 8**Regression Analysis of Total, Permanent, and Temporary Price Impacts**

	Total	Permanent	Temporary
Intercept	0.295; (0.082)	-0.560; (0.091)	1.099; (0.074)
Sign	1.007; (0.084)	1.059; (0.093)	-0.027; (0.077)
Flow/ADV	0.058; (0.006)	0.033; (0.007)	0.019; (0.005)
R-Squared	0.016	0.010	0.001
F-Statistic	156.1	96.07	5.91
Deg. Freedom	18,818	18,803	18,804

NOTES: Trade sign is defined previously in Equation 4. Flow/ADV is flow (in dollars) based on Equation 4 and divided by 20-day average daily dollar volume prior to the announcement date. We report regressions for three price impact measures as defined in Equations 1–3, measured in percentages and beta adjusted. Figures in parentheses are standard errors.

SOURCES: BlackRock as of July 31, 2021. Data on ADV are from Bloomberg.

CONCLUSIONS

Recent academic studies concluded that price impacts associated with index rebalancing can be large and may represent a hidden cost to fund investors. In this article, we discuss the practical realities of index rebalancing. Institutional context is critical; a diverse and complex ecosystem exists today to support the provision of liquidity for index rebalancing. Further, the growth of indexing means that we cannot study the price impacts of index changes in isolation.

To understand the costs of liquidity provision, we examine the return behavior around 18,871 index rebalances for the Russell indexes. A key difference from the previous literature is that we estimate net flows for each stock aggregated across all Russell indexes rather than the gross flows into each index. This is important because stocks migrating from one index to another may see positive or negative flows depending on the dollar amount of assets tracking those indexes and the

weight changes of the stock. The annual Russell reconstitution offers a unique way to understand the impact of net flows because these indexes are all reconstituted on the same day, the last Friday in June, based on objective criteria such as market capitalization. There are no other index rebalances in that week, allowing for a pure study of the cost of liquidity provision.

We argue that the correct measure of the costs paid by indexers for the supply of liquidity at a specific point in time is the beta-adjusted temporary return from the close on the effective date to a future date. In our analysis, we use three days, but our findings are robust to longer intervals of time. Done in this way, we find many instances in which costs are negative, consistent with a highly competitive market for liquidity provision. Run-ups prior to the effective date are sizeable but are not actionable and likely reflect permanent impacts associated with index changes. Whether these movements are predictable *ex ante* is a question for future research.

From a practical perspective, our results highlight the importance of looking at rebalancing in a general equilibrium versus a partial equilibrium framework, recognizing that segments of the equity market may be tracked by investors whose asset bases are quite diverse. Further, the results here point to the value of coordinating among liquidity providers and index managers. Indeed, one can imagine that efforts by index managers to trade ahead of or after a rebalance may lead to greater volatility as liquidity suppliers no longer have a focal point in time. Finally, we highlight the importance of the diversity of liquidity providers, including corporations, in achieving desirable outcomes for index investors.

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